

Virtual Private Cloud

- * VPC is a secure, isolated private cloud hosted within a public cloud provider's infrastructure.
- * VPC customers can run code, store data, host websites, & do anything else they could do in an ordinary private cloud. Hosted by a public cloud provider.
- * VPC's enable organizations to define their own private IP address space, create subnets, configure route tables & sets up network gateways.

Key features

- Security
- Control & Customization.
- Performance
- Cost Efficiency.
- Flexibility
- High Availability

Components

Subnets ⇒ A subnet is a portion of a VPC's IP range which is used to organize resources. It can be public or private for improved security & management.

- * Divides the network into smaller, manageable sections.

Routing Tables ⇒ These contain rules that define the destination for incoming & outgoing traffic.

Internet Gateways ⇒ Connects the VPC to the public internet.

* Enables communication ~~with~~ b/w instances & public IP addresses.

Types

→ NAT Gateways: Allows instances in private subnets to access the internet securely.

* Blocks inbound traffic, which ensures that the resources remain protected while receiving the updates also.

→ Virtual Private Gateways & VPN Connections

* A virtual private gateway links your VPC to on-premises networks through a secure VPN.

→ VPC Peering: - Creates private connections between multiple VPC's.

Types of Subnets

1) Public Subnet (Connects to internet via Internet Gateway).

* The public subnet has instances with public IP addresses allowing them to communicate with the internet.

Private Subnet

- * Private subnet has instances with Private IP addresses.
- * The instances cannot directly access the internet but can connect to resources within the same VPC.
- * This uses NAT gateway to get connected with Internet. (No direct route).

Use cases

<u>Public</u>	<u>Private</u>
* Web Servers,	* Databases.
* Bastion hosts.	* Internal Application servers.
* Load Balancers	* Batch Processing.

Network Security Controls (Defense in Depth)

1) Security Groups:-

- * These work at instance level.
- * Acts as a virtual Firewall for our EC2 instances.
- * These use "allow" rules.

2) Network Access Control Lists (NACLs).

* This acts as a firewall for an entire subnet.

* Known as Subnet Firewall.

Feature	Control Level.
Security Group	Instance/Resource.
Network ACL	Subnet
VPC Flow logs	Network Interface
VPC Perimeter	Service/API.

VPC Route Table

* Set of rules that determine where the network traffic from your subnet or gateway is directed.

* Each subnet is associated with a route table that controls the traffic flow b/w subnets.

* These are essential when dealing with IPv4 / IPv6 & CIDR blocks.

Components of a Route Table

- Entries.
- Route Table ID.
- Associations.
- Target
- Destination CIDR Blocks.

Types

- 1) Main Route Table.
- 2) Custom Route Table.
- 3) Gateway Route Table.

Internet Gateway

* Allows communication b/w your VPC & the internet.

* If a VPC does not have an Internet Gateway, then the resources in the VPC cannot be accessed from the internet.

* Supports IPv4 & IPv6 traffic.

NAT Gateway

* Network Address Translation.

* Allows resource in a private subnet to reach the internet without exposing them to the public.

* NAT works in one way, i.e; external services cannot connect with instances whereas internal services can connect with instances. (Outbound).

* NAT supports TCP, UDP & ICMP protocols.

* The main purpose is to translate private IP addresses to public IP's.

Load Balancer (Traffic Cop).

* This is a networking device or software application that distributes & balances the incoming traffic among the servers to provide high availability.

Problems without a load balancer

* Single point of failure $\Rightarrow$ If the server goes down then the whole application will be interrupted.

* Overloaded Servers

* Limited Scalability $\Rightarrow$ More servers cannot be added based on the work load or requirement.

A load balancer receives incoming requests, checks server health, & routes each request to the most suitable available server.

Characteristics of Load Balancer

- Traffic distribution.
- High availability.
- Scalability.
- ~~Optimal~~ Optimization.
- Health Monitoring.
- SSL Termination.

Types

1) Software Load Balancers

- * Run on general-purpose servers.
- * Application chooses the first one in the list & requests data from the server.
- * Cheapest way to implement load balancing.

2) Hardware Load Balancers

- * Uses physical appliance to distribute the traffic across the cluster of network servers.
- * These are known as Layer 4-7 Routers.
- * Handles a large volume of traffic.

Virtual Load Balancers

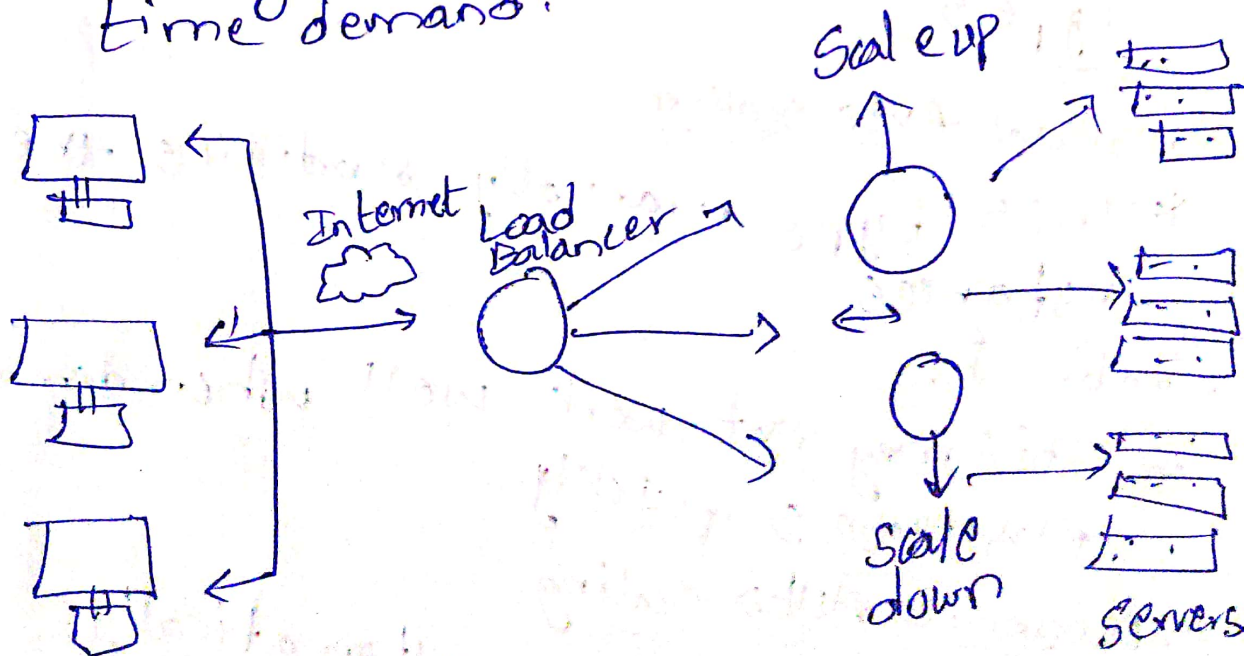
* Implemented as a Virtual Machine (VM) or software instance within a virtualized environment.

Major Examples of Load Balancers

- Ex:-
- Direct Routing Requesting Dispatching
 - Dispatcher-Based Load Balancing Cluster.
 - Linux Virtual Load Balancer.

Auto Scaling

* This is a mechanism that automatically adjusts computational resource such as virtual machines, CPU or the memory up or down based on real-time demand.



* This enables systems to effectively accommodate variations in demand.

Advantages

- Optimized Performance
- Cost Efficiency.
- Improved Reliability.
- Scalability.

Key Components

- Launch Configuration.
- Auto scaling Groups (ASG).
- Scaling Policies.
- Scaling Cooldowns.
- Health Checks.
- CloudWatch Alarms.
- Lifecycle Hooks.

Types

1) Manual Auto scaling

* This type manually modifies the number of instances.

* This may not work well when demand fluctuates quickly.

2) Dynamic Auto scaling

* Dynamic Auto scaling automatically modifies the number of instances.

* This reacts quickly to shifting workloads.

3) Predictive Auto Scaling

* Using ML algorithms & historical data, Predictive auto scaling predicts future demand & modifies resources in front of spikes.

4) Scheduling Auto Scaling

* For workloads that are predictable, like a web application that has increased traffic during business hours.
* This is based on established consumption patterns.

5) Target Tracking Auto Scaling

* This technique modifies resources in order to maintain a particular target